

GE Drives

INSTRUCTIONS

FOUR QUADRANT

PARALLEL BRIDGE

ADDENDUM TO
GEK-85767

DC-300™ Adjustable Speed Drives

™ TRADEMARK OF GENERAL ELECTRIC COMPANY, U.S.A.

INTRODUCTION

This instruction book serves as an addendum to the Installation, Operation and Maintenance Instructions provided in GE publication GEK-85767. The DC-300™ motor drive described in this addendum utilizes the two high current DC-300 drive converters interconnected to provide four quadrant (regenerative) control. Basic equipment and operation are described in GEK-85767 and additional information specific to this application is provided in the Custom Instruction Book and elementary drawings supplied with the drive.

GENERAL DESCRIPTION

The high current, four quadrant, DC-300 drive consists of two Full Wave Three Phase SCR Bridges interconnected to form a single motor drive controller as shown in Figure 1, POWER INTERCONNECTION DIAGRAM. CASE 1 houses a complete DC-300 drive controller including the Forward Bridge, Motor Field Control assembly and controller cards. Also included in this section are the AC Input Bus, Input Current Transformers (CT1 & CT3), DC Output Bus and Shunt for the drive. CASE 2 houses the Reverse Bridge which is connected to the AC Input Bus and the DC Output Bus located in CASE 1. Coordinated DC contactors (MD1 & MD2) are supplied on the output of each bridge. The drive controller provides four quadrant operation by regulating the firing of the Forward Bridge and Reverse Bridge SCRs as required.

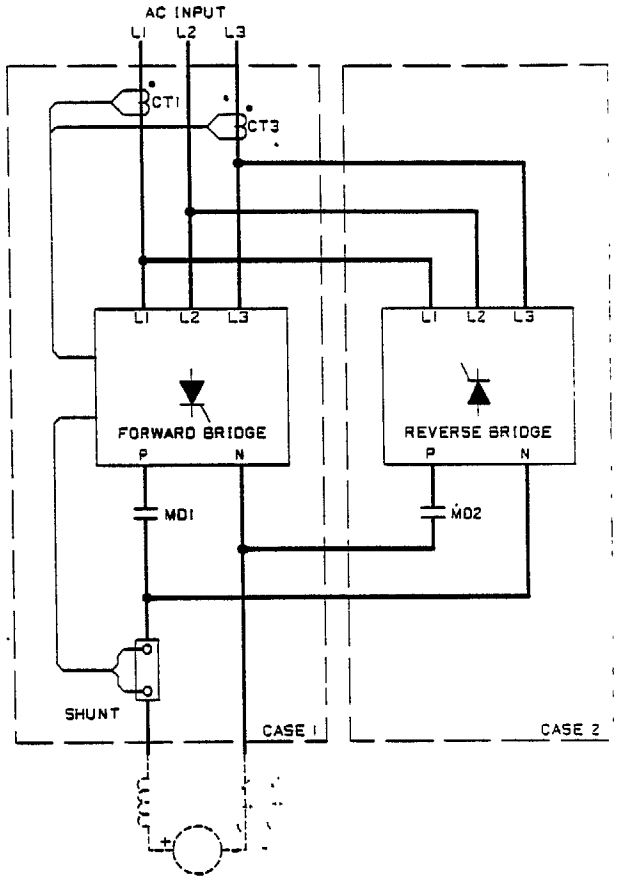


FIGURE 1
REGENERATIVE CONTROLLER — PARALLEL BRIDGE

CONTROL CIRCUITRY

Printed circuit cards, mounted in the card enclosure of the Forward Bridge, provide control for both the Forward and Reverse SCR Bridge Rectifiers: regulating the drive output in response to reference, feedback and other control signals.

SCR bridge and controller card connections are shown in FIGURE 3, CONNECTION DIAGRAM — PARALLEL FORWARD BRIDGE, and FIGURE 4, CONNECTION DIAGRAM — PARALLEL REVERSE BRIDGE. The Control, Power Supply, Power Connection and option cards are described in GEK-85767.

SCR firing signals are routed from the Control card through 5PL to the Power Connection card of the Forward Bridge. Jumpers on the Power Connection card direct the appropriate firing signals to the Forward Bridge SCRs or through connector 5APL to the Power Connection card and SCRs of the Reverse Bridge.

Ribbon cable jumper connections for the two Power Connection cards are shown in Figure 2, POWER CONNECTION CARD,. INTERCONNECTION DIAGRAM.

PARALLEL SCR BRIDGES

The Forward and Reverse SCR bridges are essentially the same, each incorporating twelve Press-Pak™ SCR cells mounted as six parallel pairs in air-cooled heatsink stack assemblies. To provide four quadrant control: the outputs of the two bridges are connected to the DC Bus in opposite polarities and the SCR firing directed to the appropriate SCR bridge for the required polarity.

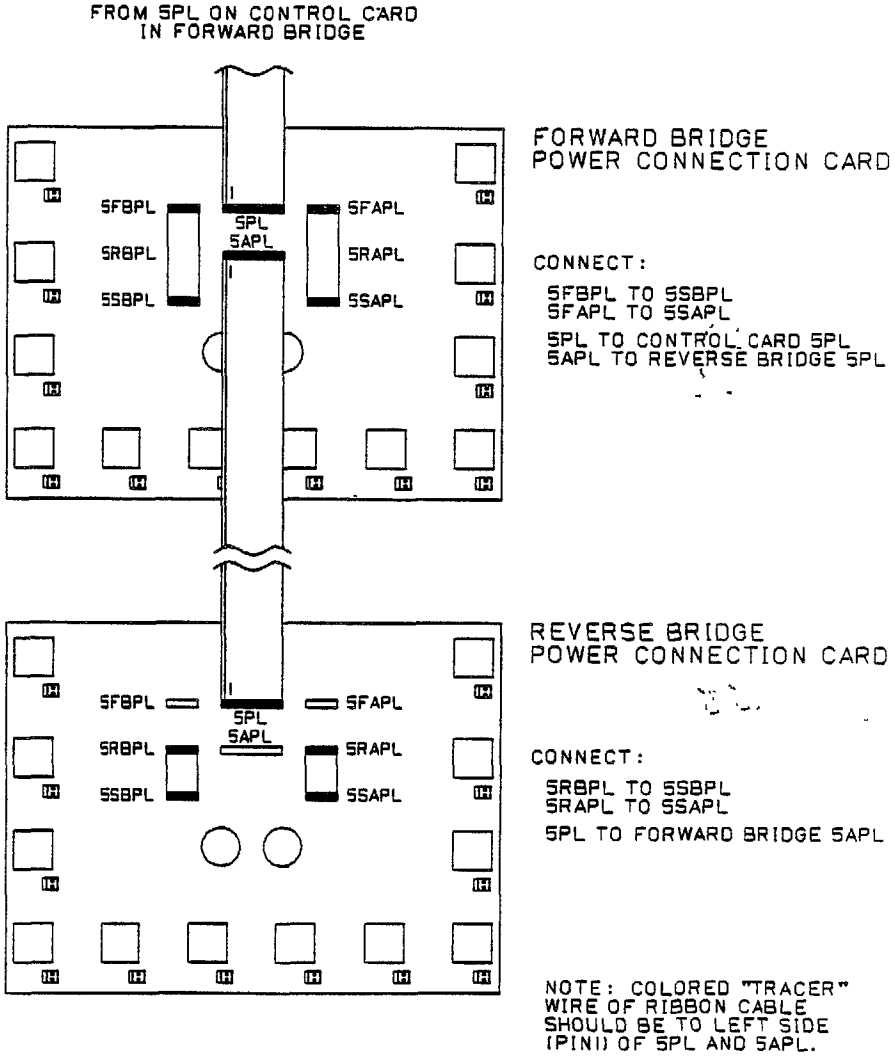


FIGURE 2
POWER CONNECTION CARD
INTERCONNECTION DIAGRAM

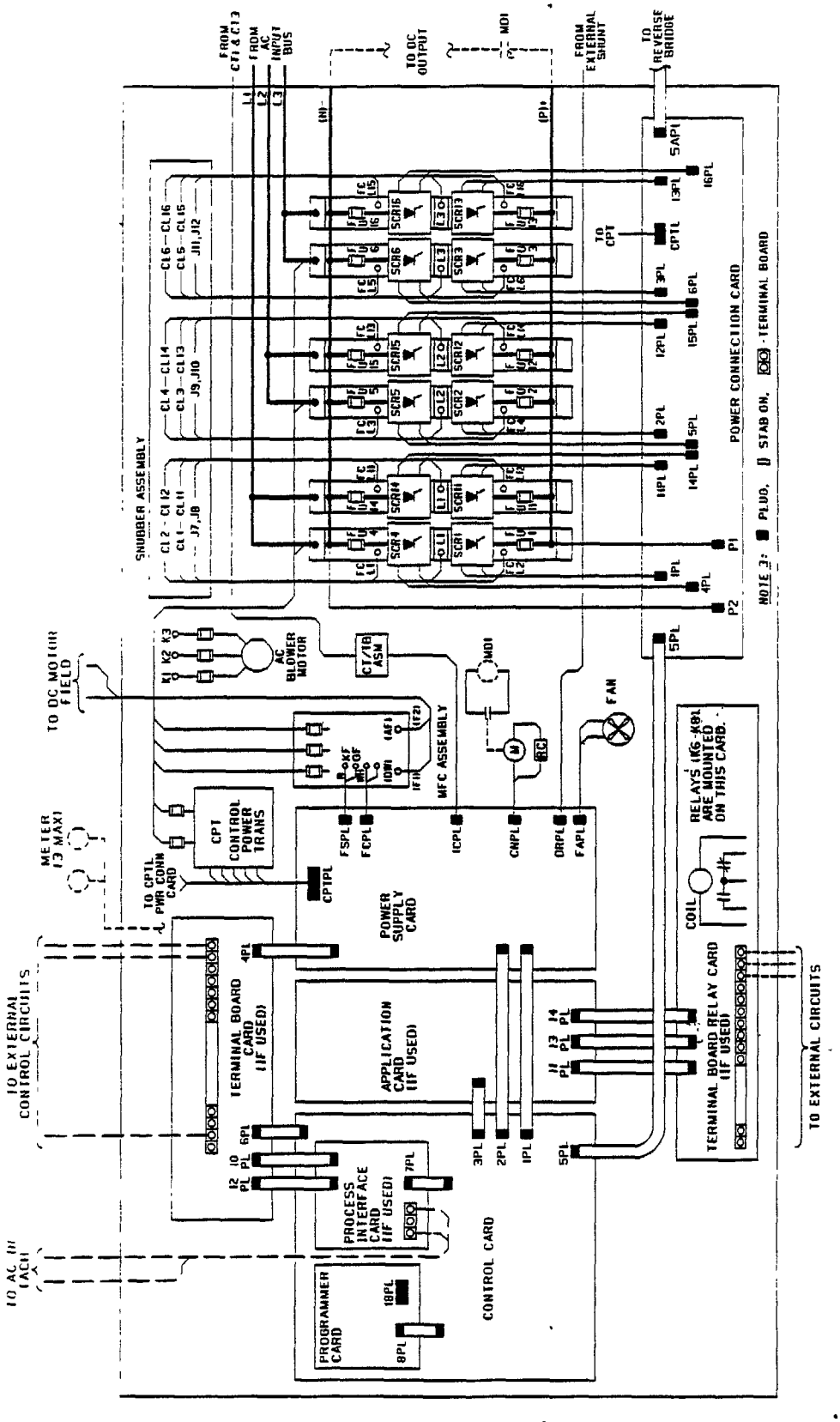


FIGURE 3
CONNECTION DIAGRAM — PARALLEL FORWARD BRIDGE

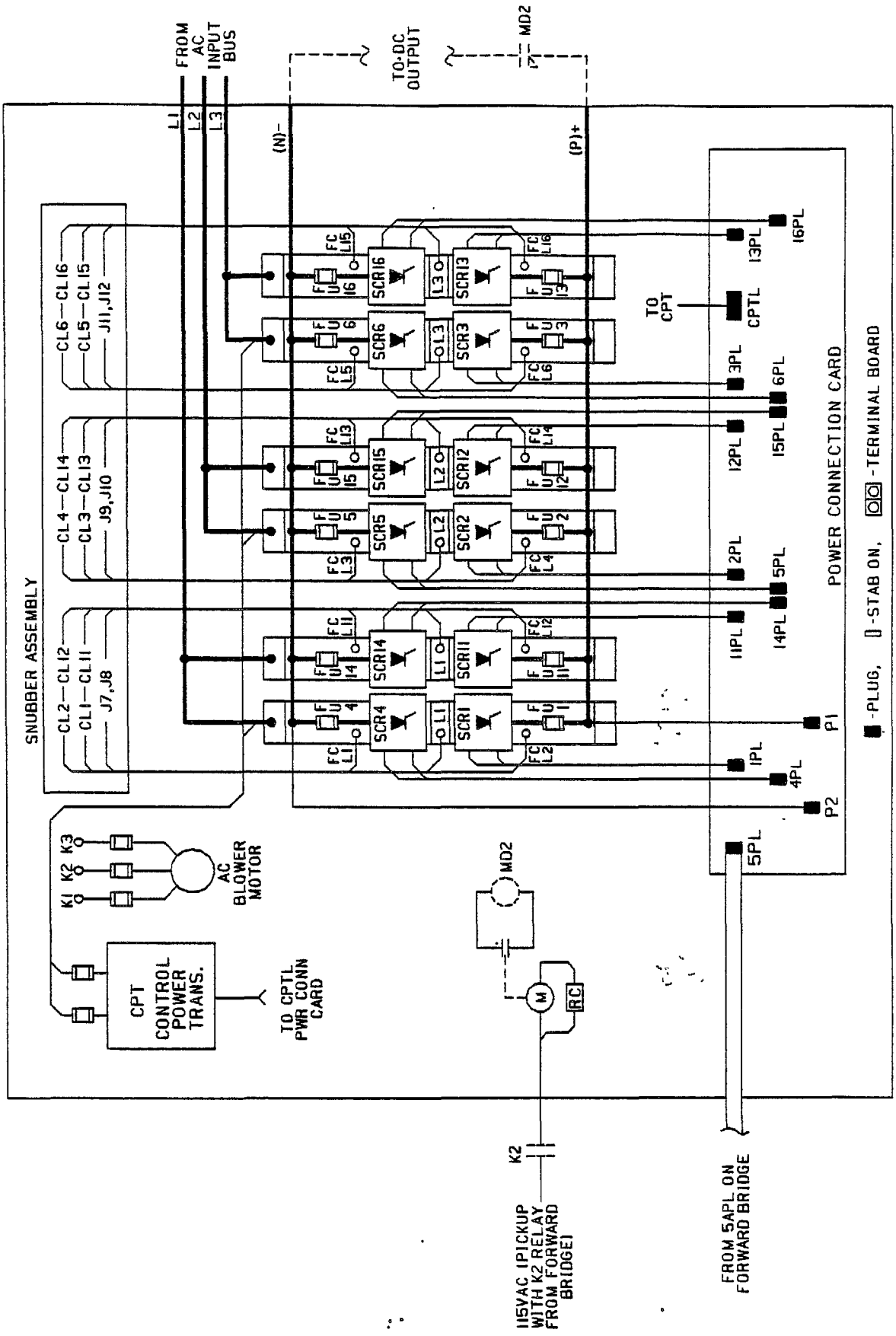


FIGURE 4
CONNECTION DIAGRAM — PARALLEL REVERSE BRIDGE



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